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Motion in a Straight Line

NEET Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	42 questions
Answer key	at the end

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MCQ (Single Correct Answer) 42 questions

Q1. RE-NEET 2026

Consider a particle moving along a straight line, whose position as a function of time is given by $s(t) = \alpha t^2 - \beta t + \gamma$, where $\alpha = 1\ \text{ms}^{-2}$, $\beta = 6\ \text{ms}^{-1}$ and $\gamma = 5\ \text{m}$. The average speed of the particle, in ms^{-1} from $t = 0$ to $t = 6\ \text{s}$ is:

Open solution

(A) 0 (B) 12 (C) 6 (D) 3

Q2. NEET 2026

The following plots show variation of velocity (v) with time (t) of a ball thrown vertically upward, and falling back. Which of the following plots is/are correct?



Open solution

(A) C only (B) D only (C) B only (D) A and E only

Q3. NEET 2026

When a ruler falls vertically, 5 different persons catch it with different reaction times. ($g = 9.8\ \text{m s}^{-2}$)

A. Person A has reaction time of 0.20 s .

B. Person B has reaction time of 0.22 s .

C. Person C has reaction time of 0.18 s .

D. Person D has reaction time of 0.19 s .

E. Person E has reaction time of 0.21 s .

What is the correct order of the distance travelled by the ruler for each person?

Open solution

(A) B > E > A > C > D (B) C > D > A > B > E

(C) B > E > A > D > C (D) C > D > A > E > B

Q4. NEET 2025

In some appropriate units, time (t) and position (x) relation of a moving particle is given by $t = x^2 + x$. The acceleration of the particle is

Open solution

(A) $+\frac{2}{(x+1)^3}$ (B) $+\frac{2}{2x+1}$ (C) $-\frac{2}{(x+2)^3}$ (D) $-\frac{2}{(2x+1)^3}$

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Q5. NEET 2025

Two cities X and Y are connected by a regular bus service with a bus leaving in either direction every T min. A girl is driving scooty with a speed of $60\ \text{km/h}$ in the direction X to Y notices that a bus goes past her every 30 minutes in the direction of her motion, and every 10 minutes in the opposite direction. Choose the correct option for the period T of the bus service and the speed (assumed constant) of the buses.

Open solution

(A) $10\ \text{min}, 90\ \text{km/h}$
(B) $15\ \text{min}, 120\ \text{km/h}$
(C) $9\ \text{min}, 40\ \text{km/h}$
(D) $25\ \text{min}, 100\ \text{km/h}$

Q6. NEET 2024 (Re-Examination)

A particle is moving along x-axis with its position (x) varying with time (t) as $x = \alpha t^4 + \beta t^2 + \gamma t + \delta$. The ratio of its initial velocity to its initial acceleration, respectively, is:

Open solution

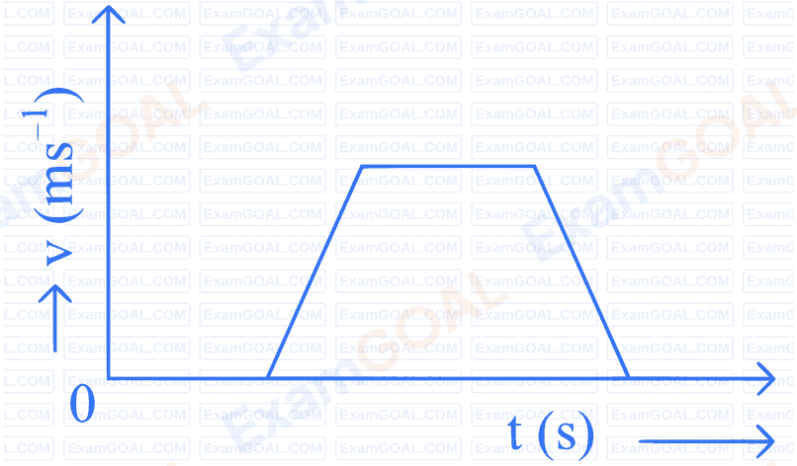
(A) $2\alpha : \delta$
(B) $\gamma : 2\delta$
(C) $4\alpha : \beta$
(D) $\gamma : 2\beta$

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Q7. NEET 2024

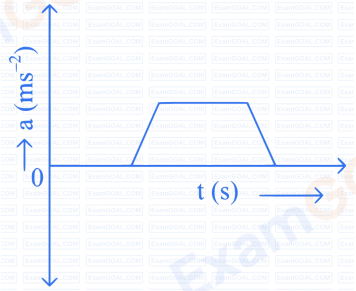
Open solution

The velocity $(v)-$ time (t) plot of the motion of a body is shown below:

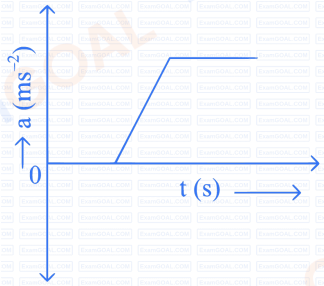


The acceleration $(a)-$ time (t) graph that best suits this motion is :

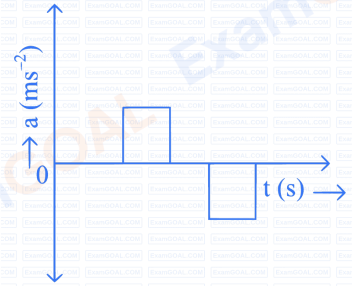
(A)



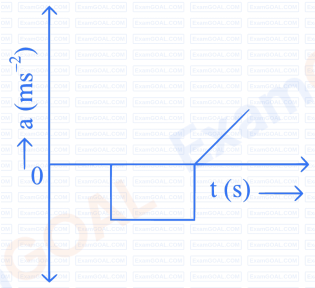
(B)



(C)



(D)



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Q8. NEET 2023 Manipur

The position of a particle is given by

$\vec{r}(t) = 4t\hat{i} + 2t^2\hat{j} + 5\hat{k}$

where t is in seconds and r in metre. Find the magnitude and direction of velocity $v(t)$, at $t = 1$ s, with respect to x-axis

Open solution

(A) $4\sqrt{2}\,\text{ms}^{-1}, 45^\circ$
(B) $4\sqrt{2}\,\text{ms}^{-1}, 60^\circ$
(C) $3\sqrt{2}\,\text{ms}^{-1}, 30^\circ$
(D) $3\sqrt{2}\,\text{ms}^{-1}, 45^\circ$

Q9. NEET 2023

A vehicle travels half the distance with speed v and the remaining distance with speed $2v$. Its average speed is :

Open solution

(A) $\frac{2v}{3}$
(B) $\frac{4v}{3}$
(C) $\frac{3v}{4}$
(D) $\frac{v}{3}$

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Q10. NEET 2022 Phase 2

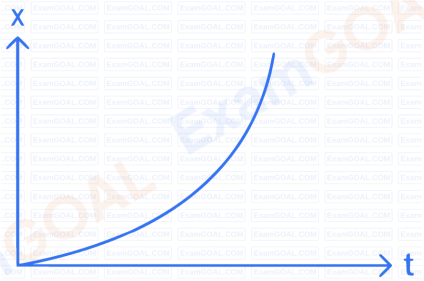
The position-time (x - t) graph for positive acceleration is

Open solution

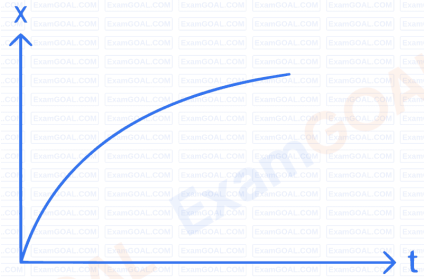
(A)



(B)



(C)



(D)



Q11. NEET 2022 Phase 1

The ratio of the distances travelled by a freely falling body in the 1st, 2nd, 3rd and 4th second

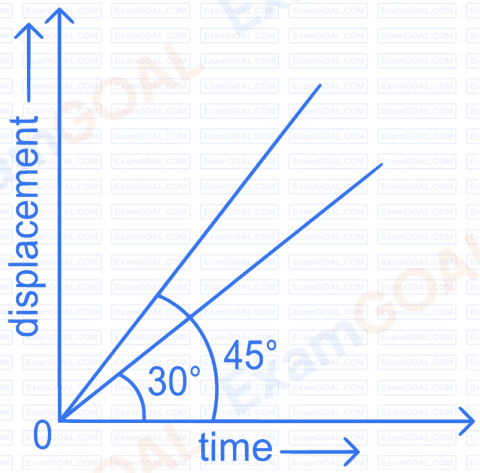
(A) 1 : 2 : 3 : 4 (B) 1 : 4 : 9 : 16 (C) 1 : 3 : 5 : 7 (D) 1 : 1 : 1 : 1

Open solution

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Q12. NEET 2022 Phase 1

The displacement-time graphs of two moving particles make angles of 30° and 45° with the x-axis as shown in the figure. The ratio of their respective velocity is



Open solution

(A) $\sqrt{3}:1$ (B) 1 : 1 (C) 1 : 2 (D) $1:\sqrt{3}$

Q13. NEET 2021

A small block slides down on a smooth inclined plane, starting from rest at time t = 0. Let S_n be the distance travelled by the block in the interval t = n − 1 to t = n. Then the ratio $\frac{S_n}{S_{n+1}}$ is :

Open solution

(A) $\frac{2n}{2n-1}$ (B) $\frac{2n-1}{2n}$ (C) $\frac{2n-1}{2n+1}$ (D) $\frac{2n+1}{2n-1}$

Q14. NEET 2020 Phase 1

A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is : (g = 10 m/s^2)

Open solution

(A) 340 m (B) 320 m (C) 300 m (D) 360 m

Q15. NEET 2018

A toy car with charge q moves on a frictionless horizontal plane surface under the influence of a uniform electric field $\vec{E}$. Due to the force q$\vec{E}$, its velocity increases from 0 to 6 m s^{-1} in one second duration. At that instant the direction of the field is reversed. The car continues to move for two more seconds under the influence of this field. The average velocity and the average speed of the toy car between 0 to 3 seconds are respectively

Open solution

(A) 2 m s^{-1}, 4 m s^{-1} (B) 1 m s^{-1}, 3 m s^{-1}
(C) 1 m s^{-1}, 3.5 m s^{-1} (D) 1.5 m s^{-1}, 3 m s^{-1}

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Q16. NEET 2017

Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1. On another day, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2. The time taken by her to walk up on the moving escalator will be

Open solution

(A) $\frac{t_1 t_2}{t_2 - t_1}$ (B) $\frac{t_1 t_2}{t_2 + t_1}$

(C) $t_1 - t_2$ (D) $\frac{t_1 + t_2}{2}$

Q17. NEET 2016 Phase 2

Two cars P and Q start from a point at the same time in a straight line and their positions are represented by

$x_P(t) = (at + bt^2)$ and $x_Q(t) = (ft - t^2)$.

Open solution

At what time do the cars have the same velocity ?

(A) $\frac{a-f}{1+b}$ (B) $\frac{a+f}{2(b-1)}$

(C) $\frac{a+f}{2(1+b)}$ (D) $\frac{f-a}{2(1+b)}$

Q18. NEET 2016 Phase 1

If the velocity of a particle is $v = At + Bt^2$, where A and B are constants, then the distance travelled by it between 1 s and 2 s is

Open solution

(A) $\frac{3}{2}A + \frac{7}{3}B$ (B) $\frac{A}{2} + \frac{B}{3}$ (C) $\frac{3}{2}A + 4B$ (D) $3A + 7B$

Q19. AIPMT 2015 Cancelled Paper

A particle of unit mass undergoes one-dimensional motion such that its velocity varies according to $v(x) = \beta x^{-2n}$, where β and n are constants and x is the position of the particle. The acceleration of the particle as a function of x, is given by

Open solution

(A) $-2\beta^2 x^{-2n+1}$ (B) $-2n\beta^2 e^{-4n+1}$ (C) $-2n\beta^2 x^{-2n-1}$ (D) $-2n\beta^2 x^{-4n-1}$

Q20. NEET 2013 (Karnataka)

The displacement 'x' (in meter) of a particle of mass 'm' (in kg) moving in one dimension under the action of a force, is related to time 't' (in sec) by $t = \sqrt{x} + 3$. The displacement of the particle when its velocity is zero, will be

Open solution

(A) 4 m (B) 0 m (zero) (C) 6 m (D) 2 m

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Q21. NEET 2013

A stone falls freely under gravity. It covers distances h_1, h_2 and h_3 in the first 5 seconds, the next 5 seconds and the next 5 seconds respectively. The relation between h_1, h_2 and h_3 is

(A) $h_2 = 3h_1$ and $h_3 = 3h_2$ (B) $h_1 = h_2 = h_3$

(C) $h_1 = 2h_2 = 3h_3$ (D) $h_1 = \frac{h_2}{3} = \frac{h_3}{5}$

Open solution

Q22. AIPMT 2012 Prelims

The motion of a particle along a straight line is described by equation $x = 8 + 12t - t^3$ where x is in metre and t in second. The retardation of the particle when its velocity becomes zero is

(A) 24 m s^{-2} (B) zero (C) 6 m s^{-2} (D) 12 m s^{-2}

Open solution

Q23. AIPMT 2011 Mains

A particle covers half of its total distance with speed v_1 and the rest half distance with speed v_2. Its average speed during the complete journey is

(A) $\frac{v_1+v_2}{2}$ (B) $\frac{v_1v_2}{v_1+v_2}$

(C) $\frac{2v_1v_2}{v_1+v_2}$ (D) $\frac{v_1^2v_2^2}{v_1^2+v_2^2}$

Open solution

Q24. AIPMT 2011 Prelims

A boy standing at the top of a tower of 20 m height drops a stone. Assuming g = 10 m s^{-2}, the velocity with which it hits the ground is

(A) 10.0 m/s (B) 20.0 m/s (C) 40.0 m/s (D) 5.0 m/s

Open solution

Q25. AIPMT 2010 Prelims

A particle moves a distance x in time t according to equation $x = (t + 5)^{-1}$. The acceleration of particle is proportional to

(A) $(\text{velocity})^{3/2}$ (B) $(\text{distance})^2$ (C) $(\text{distance})^{-2}$ (D) $(\text{velocity})^{2/3}$

Open solution

Q26. AIPMT 2010 Prelims

A ball is dropped from a high rise platform at t = 0 starting from rest. After 6 seconds another ball is thrown downwards from the same platform with a speed v. The two balls meet at t = 18 s. What is the value of v?

(Take g = 10 m/s^2)

(A) 75 m/s (B) 55 m/s (C) 40 m/s (D) 60 m/s

Open solution

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Q27. AIPMT 2009

A particle starts its motion from rest under the action of a constant force. If the distance covered in first 10 seconds is S_1 and that covered in the first 20 seconds is S_2, then

(A) $S_2 = 3S_1$ (B) $S_2 = 4S_1$ (C) $S_2 = S_1$ (D) $S_2 = 2S_1$

Open solution

Q28. AIPMT 2009

A bus is moving with a speed of 10 ms^{-1} on a straight road. A scooterist wishes to overtake the bus in 100 s. If the bus is at a distance of 1 km from the scooterist, with what speed should the scooterist chase the bus ?

(A) 40 m s^{-1} (B) 25 m s^{-1} (C) 10 m s^{-1} (D) 20 m s^{-1}

Open solution

Q29. AIPMT 2008

A particle moves in a straight line with a constant acceleration. It changes its velocity from 10 ms^{-1} to 20 ms^{-1} while passing through a distance 135 m in t second. The value of t is

(A) 12 (B) 9 (C) 10 (D) 1.8

Open solution

Q30. AIPMT 2008

The distance travelled by a particle starting from rest and moving with an acceleration $\frac{4}{3}$m s^{-2}, in the third second is

(A) $\frac{10}{3}$ m (B) $\frac{19}{3}$ m (C) 6 m (D) 4 m

Open solution

Q31. AIPMT 2007

A particle moving along x-axis has acceleration f, at time t, given by $f = f_0\left(1 - \frac{t}{T}\right)$, where f_0 and T are constants. The particle at t = 0 has zero velocity. In the time interval between t = 0 and the instant when f = 0, the particle's velocity (v_x) is

(A) $\frac{1}{2} f_0T^2$ (B) f_0T^2 (C) $\frac{1}{2} f_0T$ (D) f_0T

Open solution

Q32. AIPMT 2007

A car moves from X to Y with a uniform speed v_u and returns to X with a uniform speed v_d. The average speed for this round trip is

(A) $\sqrt{v_u v_d}$ (B) $\frac{v_d v_u}{v_d + v_u}$

(C) $\frac{v_u + v_d}{2}$ (D) $\frac{2v_d v_u}{v_d + v_u}$

Open solution

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Q33. AIPMT 2007

The position x of a particle with respect to time t along x-axis is given by $x = 9t^2 - t^3$ where x is in metres and t in seconds. What will be the position of this particle when it achieves maximum speed along the + x direction ?

(A) 54 m (B) 81 m (C) 24 m (D) 32 m

Open solution

Q34. AIPMT 2006

Two bodies A (of mass 1 kg) and B (of mass 3 kg) are dropped from heights of 16 m and 25 m, respectively. The ratio of the time taken by them to reach the ground is

(A) 4/5 (B) 5/4 (C) 12/5 (D) 5/12

Open solution

Q35. AIPMT 2006

A car runs at a constant speed on a circular track of radius 100 m, taking 62.8 seconds for every circular lap. The average velocity and average speed for each circular lap respectively is

(A) 10 m/s, 0 (B) 0, 0 (C) 0, 10 m/s (D) 10 m/s, 10 m/s.

Open solution

Q36. AIPMT 2006

A particle moves along a straight line OX. At a time t (in seconds) the distance x (in metres) of the particle from O is given by $x = 40 + 12t - t^3$. How long would the particle travel before coming to rest ?

(A) 16 m (B) 24 m (C) 40 m (D) 56 m

Open solution

Q37. AIPMT 2005

A ball is thrown vertically upward. It has a speed of 10 m/sec when it has reached one half of its maximum height. How high does the ball rise?
(Take $g = 10\ m/s^2$.)

(A) 10 m (B) 5 m (C) 15 m (D) 20 m

Open solution

Q38. AIPMT 2005

The displacement x of a particle varies with time t as $x = ae^{-\alpha t} + be^{\beta t}$, where a, b, α and β are positive constants. The velocity of the particle will

(A) be independent of β (B) drop to zero when $\alpha = \beta$

(C) go on decreasing with time (D) go on increasing with time.

Open solution

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Q39. AIPMT 2003

A man throws balls with the same speed vertically upwards one after the other at an interval of 2 seconds. What should be the speed of the throw so that more than two balls are in the sky at any time ? (Given g = 9.8 m/s^2)

Open solution

(A) more than 19.6 m/s
(B) at least 9.8 m/s
(C) any speed less than 19.6 m/s
(D) only with speed 19.6 m/s.

Q40. AIPMT 2003

If a ball is thrown vertically upwards with speed u, the distance covered during the last t seconds of its ascent is

Open solution

(A) ut
(B) $\frac{1}{2}$ gt^2
(C) $ut - \frac{1}{2} gt^2$
(D) (u + gt) t.

Q41. AIPMT 2001

A particle is thrown vertically upward. Its velocity at half of the height is 10 m/s, then the maximum height attained by it (g = 10 m/s^2)

Open solution

(A) 8 m
(B) 20 m
(C) 10 m
(D) 16 m

Q42. AIPMT 2000

Motion of a particle is given by equation s = ($3t^3 + 7t^2 + 14t + 8$) m. The value of acceleration of the particle at t = 1 sec is

Open solution

(A) 10 m/s^2
(B) 32 m/s^2
(C) 23 m/s^2
(D) 16 m/s^2

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Answer Key

1. **D**	2. **A**	3. **C**	4. **D**	5. **B**
6. **D**	7. **C**	8. **A**	9. **B**	10. **B**
11. **C**	12. **D**	13. **C**	14. **C**	15. **B**
16. **B**	17. **D**	18. **A**	19. **D**	20. **B**
21. **D**	22. **D**	23. **C**	24. **B**	25. **A**
26. **A**	27. **B**	28. **D**	29. **B**	30. **A**
31. **C**	32. **D**	33. **A**	34. **A**	35. **C**
36. **A**	37. **A**	38. **D**	39. **A**	40. **B**
41. **C**	42. **B**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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